

Reina Servano

831-920-8721 Davis, CA rservano@ucdavis.edu <https://www.linkedin.com/in/reinaservano/>

EDUCATION

University of California, Davis

June 2028

Bachelor of Science in Electrical Engineering

Coursework: Circuits I&II, Introductory Electromagnetics I&II, Digital Signals & Systems, Digital Systems I

WORK EXPERIENCE

Hardware Lead

Davis, CA

Northwest AI Hub and Robert Mondavi Institute Winery

June 2026 - Present

- Designing and documenting schematics that support the Software Lead's Python program for the Dual-Axis Barrel Scanner, connecting the two motors, the Rotation Axis Motor and the Tilt Axis Motor, to the dedicated control PC and LCD display
- Assembling electrical connections powering the 24VDC components including the STEPPER MOTOR IDM57 and ESS17, soldering and securing wires, and crafting 3D-printed mounts using MCAD, while focusing on the safety of the Winemakers
- Conducting verification and validation, testing that the wiring and hardware for the Dual-Axis Barrel Scanner safe, efficient, and is capable of accurately scanning the wine barrels with the automated motors talking to the control PC through the RS485 cable

Embedded Security Summer Program Teacher Assistant

Cambridge, MA

Massachusetts Institute of Technology

July 2024 - August 2024

- Developed 2 qualitative Jupyter notebook labs for the Embedded Systems course, with an emphasis on Hardware Hacking, improving 20 students' mastery in cryptographic techniques such as Diffie-Hellman and Hashing
- Enhanced student comprehension of authentication and encryption by writing a student reference guide for the WolfSSL Library, allowing students to efficiently navigate and apply the library in their hands-on firmware development project
- Assisted in testing and validating the students' MITRE-CAR Ti-MCU board project, helping ensure successful programming, building, and flashing of embedded code while working with a newly introduced SSL Library

ACTIVITIES

Firmware Lead and VP of Tech

October 2025 - June 2026

UC Davis Google Developers Students Club

- Lead a firmware-focused project of 4 developers, working on an embedded systems project for solving engineering problems in the community, and expanding hands-on computer engineering experience on campus
- Lead a team of 8 Technical Directors who are leading various 8 firmware, hardware, and software projects, including but not limited to AI/ML, ensuring that developers are equipped with successful resources and workshops for their projects

Research Proposal Writer and Evaluator

January 2026 - May 2026

NASA L'Space Proposal Writing and Evaluation Experience

- Assisted with authoring a research proposal for an autonomous spacecraft monitoring system, and developing NASA research ideation charts, while attending meetings and receiving training in writing proposals
- Reviewed and evaluated three NASA research proposals on computer vision for Lunar landing control, autonomy-AI, and autonomous deep subsurface exploration, while receiving training in proposal reviews

Deputy Project Manager of Resources

September 2025 - December 2025

NASA L'Space Mission Concept Academy

- Organized the logistics team of the Black Widow Martian cave exploration mission concept, leading 4 student mission assurance and program analysts, ensuring productivity and cooperation in completing the mission design deliverables
- Authored programmatic summaries of the Preliminary Design Review, detailing the mission cost and schedule overview to assure mission feasibility, mitigating controllable financial and timely risks

PROJECTS

One-Hot Encoded Dice Game

- Designed and programmed an FPGA-based dice game of rolling 22, utilizing clocked and clock-enabled Flip-Flops, a custom-made binary to hexadecimal display decoder, original score tracker-comparator, and game turn counter

SKILLS AND INTERESTS

Software: HFSS, SPICE, Altium Designer, Ki-CAD, FPGA, Excel, Autodesk Fusion

Technical: Wiring, Soldering, Keysight Oscilloscopes, Bambu Lab 3D Printing, Arduino

Programming: C, C++, Python, Matlab, Git, UNIX, Docker, PyTorch, TensorFlow

Interests: RF, High-Frequency Waves, Analog Electronics, Mixed-Signal IC Design, Electromagnetics